

TOC-2026: Quiz-6 (Reduction Procedure and Undecidability)

Full Marks: 20, Duration: 40 minutes

(1) Consider the language $A_{TM} = \{\langle M, w \rangle \mid M \text{ is a Turing-Machine and } M \text{ accepts } w\}$. We describe following procedure.

On input $\langle M, w \rangle$:

(i) Construct Turing-Machine M_1 as follows

M_1 on input x

- If $(x \notin \{0, 1\}^*)$, then REJECT;
- Else ACCEPT;

(ii) Construct Turing-Machine M_2 as follows

M_2 on input y

- If $(y \neq 01)$ then REJECT;
- Run M on w ;
- If $(M \text{ accepts } w)$ then ACCEPT;
- If $(M \text{ does not accept } w)$ then REJECT;

(ii) Output $\langle M_1, M_2 \rangle$;

Consider the language $BTM = \{\langle M_1, M_2 \rangle \mid M_1 \text{ and } M_2 \text{ are Turing-Machines such that } L(M_1) \cdot L(M_2) \text{ is the set of all strings over } \{0, 1\}^* \text{ that end with } 01\}$.

Formally prove that $\langle M, w \rangle \in A_{TM}$ if and only if $\langle M_1, M_2 \rangle \in BTM$.

(6 Marks)

SOLUTION

We prove that $\langle M, w \rangle \in A_{TM} \iff \langle M_1, M_2 \rangle \in BTM$.

($\Rightarrow$) Assume $\langle M, w \rangle \in A_{TM}$.

Then M accepts w .

By construction, M_1 accepts every string in $\{0, 1\}^*$;

Hence $L(M_1) = \{0, 1\}^*$.

By construction, M_2 rejects any input $y \neq 01$ and since M accepts w so M_2 accepts $y = 01$
Thus $L(M_2) = \{01\}$.

Therefore, $L(M_1) \cdot L(M_2) = \{0,1\}^* \cdot \{01\} = \{x01 \mid x \in \{0,1\}^*\}$,
which is exactly the set of all strings over $\{0,1\}^*$ ending with 01.
Hence $\langle M_1, M_2 \rangle \in BTM$.

($\Leftarrow$) Assume $\langle M_1, M_2 \rangle \in BTM$.

Then $L(M_1) \cdot L(M_2)$ is the set of all strings over $\{0,1\}^*$ that end with 01.

From the construction, $L(M_1) = \{0,1\}^*$ always.

Assume to the contrary that M does not accept w ,

then M_2 rejects all inputs, so $L(M_2) = \emptyset$.

Then $L(M_1) \cdot L(M_2) = \{0,1\}^* \cdot \emptyset = \emptyset$, which is not the set of strings ending with 01.

This is a contradiction.

Hence M must accept w , so that $L(M_2) = \{01\}$.

Thus $\langle M, w \rangle \in A_{TM}$.

Hence

$\langle M, w \rangle \in A_{TM} \iff \langle M_1, M_2 \rangle \in BTM$.

(2) Consider the language $SPLReg_{TM} = \{\langle M \rangle \mid M \text{ is a Turing-Machine such that } L(M) \text{ accepts at least one string over } \{0,1\}^* \text{ that starts with 11 and ends with a 0}\}$. Answer the following questions.

- (a) State what property of recursively enumerable languages does $SPLReg_{TM}$ asks for.
- (b) Explain why is the property stated in part-(a) is a nontrivial property.

(1 Marks + 6 Marks)

SOLUTION

(a) The language $SPLReg_{TM}$ asks for:

“ $\exists x \in L(M)$ such that $x \in \{0,1\}^*$, x starts with 11 and ends with 0.”

Alternatively, “ $L(M) \cap \{x \in \{0,1\}^* \mid x \text{ starts with 11 and ends with 0}\} \neq \emptyset$ ”

(b) To show that this property is nontrivial, we must show that:

- (i) there exists a TM whose language satisfies the property, and
- (ii) there exists a TM whose language does not satisfy the property.

For (i), Consider a TM M_1 such that $L(M_1) = \{110\}$.
Clearly, 110 starts with 11 and ends with 0, hence $L(M_1)$ satisfies the property.

For (ii), Consider a TM M_2 such that $L(M_2) = \emptyset$
Then $L(M_2)$ contains no string that starts with 11 and ends with 0, hence it does not satisfy the property.

Hence, the property is nontrivial.

(3) Consider the language $STM = \{\langle M \rangle \mid M \text{ is a Turing-Machine such that } L(M) \text{ contains at least 3 strings}\}$. Provide a precise mapping reduction (with proof) from A_{TM} to STM . Your reduction must contain the following two things. **(7 Marks)**

- A precise procedure that takes an instance $\langle M, w \rangle$ of A_{TM} and outputs an instance $\langle M^* \rangle$ of STM .
- Formal proof that $\langle M, w \rangle \in A_{TM}$ if and only if $\langle M^* \rangle \in STM$.

SOLUTION

Reduction:

1. On input $\langle M, w \rangle$, construct a TM M^* as follows.

M^* on input x

- If $(x \notin \{0, 1\}^*)$, then REJECT;
- Simulate M on w ;
- If M accepts w , ACCEPT;
- Else REJECT;

2. Output $\langle M^* \rangle$.

Proof of correctness: We show that $\langle M, w \rangle \in A_{TM} \iff \langle M^* \rangle \in STM$.

$(\Rightarrow)$ Assume $\langle M, w \rangle \in A_{TM}$.
Then M accepts w .

From the construction, if M accepts w , then M^* accepts all strings in $\{0, 1\}^*$.
Thus $L(M^*) = \{0, 1\}^*$, hence it contains at least 3 strings.
Therefore $\langle M^* \rangle \in STM$.

$(\Leftarrow)$ Assume $\langle M^* \rangle \in STM$.
Then $L(M^*)$ contains at least 3 strings.

Assume to the contrary that M does not accept w , then M^* simulates M on w which either loops or rejects. As a result, M^* does not accept any string

$L(M^*) = \emptyset$, which has strictly less than 3 strings.

This is a contradiction.

M must accept w .

Hence $\langle M, w \rangle \in A_{TM}$.

Hence:

$\langle M, w \rangle \in A_{TM} \iff \langle M^* \rangle \in STM.$